

Anindya Sarkar

Ph.D. Candidate, Computer Science & Engineering
Washington University in St. Louis
Advisor: Prof. Yevgeniy Vorobeychik

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RESEARCH

My goal is to build AI models that can understand, generate, plan, and reason over high-dimensional data across modalities. I am particularly interested in developing generative modeling algorithms that are mathematically grounded and scale to achieve state-of-the-art performance in data-constrained settings, such as scientific discovery and medical diagnosis. My current work aims to improve generative models' search capabilities to accelerate target discovery. I also investigate the capabilities of generative models to tackle planning and reasoning in complex, partially observable environments such as overhead and underwater environments. To tackle this broad range of problems, I draw on tools from Generative Modeling and Sampling, Applied Stochastic Processes and Control, Optimal Transport, and Reinforcement Learning.

Generative Modeling & Sampling Enhancing inference-time search and fine-grained alignment capabilities of generative models: IMPFM [Under Review], BFMT [Under Review], BSIT [Under Review], PAPA [ECML 2026], LATS [Under Review]

Applied Stochastic Process & Control Active target discovery under partially observable environments via uncertainty-aware generative models: EM-PTDM [NeurIPS 2025], DiffATD [NeurIPS 2025]

Reinforcement Learning Visual & Geospatial Active Search for social good: OWL-GPS [Under Review], VAS [WACV 2024], PS-VAS [NeurIPS 2023], GOMAA-Geo [NeurIPS 2024], DiffVAS [AAMAS 2026], AGS [AAAI 2025, AAMAS 2024], Multi-Task RL: PACMAN [ICML 2025]

Optimal Transport Out-of-distribution active target discovery via Wasserstein Gradient Flow. [In Progress]

Representation Learning Robustness & Explainability: Ane-hoc Explainability [CVPR 2022], Attributional Robustness [AAAI 2021], Adversarial Robustness [NeurIPS 2021], Robustness against Unseen Noise [WACV 2022], Reward-delay attack on RL [GameSec 2022]; Graph Neural Network: Representation Capability of Message Passing GNN [NeurIPS 2022]

EDUCATION

Sept 2021 – May 2027 (exp.) **Ph.D., Computer Science & Engineering**
Washington University in St. Louis, MO — Advisor: Prof. Yevgeniy Vorobeychik

Sept 2021 – June 2023 **M.S., Computer Science**
Washington University in St. Louis, MO

Sept 2014 – Dec 2016 **M.Tech., Bio-Medical Engineering**
Indian Institute of Technology Hyderabad, India

RESEARCH EXPERIENCE

Sept 2021 – Present **Graduate Research Assistant — Washington University in St. Louis**
Enhancing Search Capabilities of Generative Models; Uncertainty-aware generative model for active target discovery in partially observable environments; Generative Models for planning under partially observable environments; Visual & Geo-spatial Active Search. Advisor: Prof. Yevgeniy Vorobeychik.

2020 – 2021 **Research Assistant — Indian Institute of Technology Hyderabad**
Adversarial machine learning: attributional and adversarial robustness, ante-hoc explainable model architectures, test-time robustness to unseen noise. Advisor: Prof. Vineeth N. Balasubramanian.

July 2016 – Mar 2020 **R&D Engineer, Deep Learning — Quest Global Pvt. Ltd., India**
Deep learning for computer vision and natural language processing across applied products.

PUBLICATIONS

CONFERENCE PUBLICATIONS

ECML PKDD

2026

PAPA: Online Personalized Active Preference Alignment

Anindya Sarkar, Nasik Muhammad Nafi, Isaac Lyngaas, Muralikrishnan Gopalakrishnan Meena, Yevgeniy Vorobeychik. [arXiv](#)

AAMAS

2026

DiffVAS: Diffusion-Guided Visual Active Search in Partially Observable Environments

Anindya Sarkar, Srikumar Sastry, Aleksis Pirinen, Nathan Jacobs, Yevgeniy Vorobeychik. [arXiv](#)

NeurIPS

2025

Active Target Discovery under Uninformative Prior: The Power of Permanent and Transient Memory

Anindya Sarkar, Binglin Ji, Yevgeniy Vorobeychik. [arXiv](#)

NeurIPS

2025

Online Feedback Efficient Active Target Discovery in Partially Observable Environments

Anindya Sarkar, Binglin Ji, Yevgeniy Vorobeychik. [arXiv](#)

ICML

2025

Learning Policy Committees for Effective Personalization in MDPs with Diverse Tasks

Luise Ge, Michael Lanier, Anindya Sarkar, Bengisu Guresti, Chongjie Zhang, Yevgeniy Vorobeychik. [arXiv](#)

AAAI

2025

Active Geospatial Search for Efficient Tenant Eviction Outreach

Anindya Sarkar, Alex DiChristofano, Sanmay Das, Patrick Fowler, Nathan Jacobs, Yevgeniy Vorobeychik. [arXiv](#)

NeurIPS

2024

GOMAA-Geo: GOal Modality Agnostic Active Geo-localization

Anindya Sarkar, Srikumar Sastry, Aleksis Pirinen, Chongjie Zhang, Nathan Jacobs, Yevgeniy Vorobeychik. [arXiv](#)

WACV

2024

A Visual Active Search Framework for Geospatial Exploration

Anindya Sarkar, Michael Lanier, Scott Alfeld, Jiarui Feng, Roman Garnett, Nathan Jacobs, Yevgeniy Vorobeychik. [arXiv](#)

AAMAS

2024

Geospatial Active Search for Preventing Evictions

Anindya Sarkar, Alex DiChristofano, Sanmay Das, Patrick Fowler, Nathan Jacobs, Yevgeniy Vorobeychik. [ACM DL](#)

NeurIPS

2023

A Partially-Supervised Reinforcement Learning Framework for Visual Active Search

Anindya Sarkar, Nathan Jacobs, Yevgeniy Vorobeychik. [Proceedings](#)

NeurIPS

2022

How Powerful are K-hop Message Passing Graph Neural Networks

Jiarui Feng, Yixin Chen, Fuhai Li, Anindya Sarkar, Muhan Zhang. [arXiv](#)

CVPR

2022

A Framework for Learning Ante-hoc Explainable Models via Concepts

Anirban Sarkar, Deepak Vijaykeerthy, Anindya Sarkar, Vineeth N. Balasubramanian. [IEEE](#)

WACV

2022

Leveraging Test-Time Consensus Prediction for Robustness against Unseen Noise

Anindya Sarkar, Anirban Sarkar, Vineeth N. Balasubramanian. [CVF](#)

GameSec

2022

Reward Delay Attacks on Deep Reinforcement Learning

Anindya Sarkar, Jiarui Feng, Yevgeniy Vorobeychik, Christopher Gill, Ning Zhang. [arXiv](#)

NeurIPS

2021

Get Fooled for the Right Reason: Improving Adversarial Robustness through a Teacher-Guided Curriculum Learning Approach

Anindya Sarkar, Anirban Sarkar, Sowrya Gali, Vineeth N. Balasubramanian. [arXiv](#)

AAAI

2021

Enhanced Regularizers for Attributional Robustness

Anindya Sarkar, Anirban Sarkar, Vineeth N. Balasubramanian. [AAAI](#)

ICANN

2020

Enforcing Linearity in DNN Succours Robustness and Adversarial Image Generation

Anindya Sarkar, Raghu Iyengar. [Springer](#)

ICMLA

2020

ODE-Guided Neural Data Augmentation Techniques for Time Series Data and its Benefits on Robustness

Anindya Sarkar, Anirudh Raj, Raghu Iyengar. [arXiv](#)

NLPIR

2019

Zero-Shot Multilingual Sentiment Analysis using Hierarchical Attentive Network and BERT

Anindya Sarkar, Sujeeth Reddy, Raghu Iyengar. [ACM DL](#)

MANUSCRIPTS UNDER REVIEW

arXiv

2026

Sequentially-Controlled Interactive Multi-Particle Flow-Maps for Online Feedback-Driven Search

Binglin Ji, Anindya Sarkar, Hengchang Lu, Jens Sjölund, Yevgeniy Vorobeychik. [arXiv:2607.01144](#). [arXiv](#)

arXiv

2026

Bootstrap Flow-Map Tree Sampling Enables Online Feedback Driven Search

Binglin Ji, Anindya Sarkar, Hengchang Lu, Jens Sjölund, Yevgeniy Vorobeychik. [arXiv:2607.02915](#). [arXiv](#)

Under review

LATS: Lévy-Adaptive Tree Sampling

Binglin Ji, Anindya Sarkar, Hengchang Lu, Lecheng Kong, Yixin Chen, Yevgeniy Vorobeychik.

Under review

BSIT: Bootstrap Stochastic Interpolant Tree for Efficient Search under Sparse Reward

Binglin Ji, Anindya Sarkar, Yevgeniy Vorobeychik.

WORKSHOP PUBLICATIONS

AAAI Workshop

2024

Attacks on Node Attributes in Graph Neural Networks

Xu Ying, Michael Lanier, Anindya Sarkar, Yevgeniy Vorobeychik. *AI for Cyber Security*. [arXiv](#)

ECCV Workshop

2022

Empowering a Robust Model with Stable and Object-Aligned Explanations

Sowrya Gali, Anindya Sarkar, Vineeth N. Balasubramanian. *Adversarial Robustness in the Real World*. [PDF](#)

ECCV Workshop

2020

Inducing Semantic Grouping of Latent Concepts for Explanations: An Ante-Hoc Approach

Anirban Sarkar, Deepak Vijaykeerthy, Anindya Sarkar, Vineeth N. Balasubramanian. [arXiv](#)

JOURNAL PUBLICATIONS

Front. Neuroinf.

2019

Curated Model Development Using NEUROiD: A Web-Based NEUROMotor Integration and Design Platform

Anindya Sarkar. *Frontiers in Neuroinformatics*. [Link](#)

TALKS AND PRESENTATIONS

Oct 2025

Visual Active Search for Social Good

Invited talk, University of Michigan — hosted by Prof. Elizabeth Bondi-Kelly

May 2024

Doctoral Consortium — AAMAS 2024

23rd International Conference on Autonomous Agents and Multiagent Systems

Conference presentations

Presented work at **NeurIPS** (2021, 2023, 2024, 2025 $\times 2$), **AAAI** (2021, 2025), **AAMAS** (2024, 2026), **WACV** (2022, 2024), **ECML PKDD** (2026), **GameSec** (2022), **ICANN** (2020), **ICMLA** (2020), and **NLPIR** (2019).

TEACHING

Fall 2024

Teaching Assistant — CS755, Adversarial Machine Learning

Computer Science & Engineering, Washington University in St. Louis

MENTORING

2025 – 2026	Binglin Ji — M.S., Washington University in St. Louis Two NeurIPS 2025 papers (DiffATD; EM-PTDM) and four manuscripts under review on the search capabilities of generative models.
2026	Xinyu Wang — M.S., Washington University in St. Louis Safe active search in underwater environments (in progress).
2025 – 2026	Hengchang Lu — B.S., Washington University in St. Louis Three manuscripts under review on the search capabilities of generative models.
2021	Sowrya Gali — Undergraduate, Indian Institute of Technology Hyderabad [NeurIPS 2021] paper on teacher-guided curriculum learning for adversarial robustness; [ECCV Workshop 2022] paper on object-aligned robust explanation of neural network.

PROFESSIONAL SERVICE

Program committee	AAAI — Program Committee member, 2024–2026.
Conference reviewer	NeurIPS (2021–2026), ICML (2021–2026), CVPR (2022–2026), AAMAS (2022–2026), ICLR (2023–2026), ECCV (2024–2026).

HONORS AND AWARDS

2024-2026	Departmental Honors — top 15% of Ph.D. students Department of Computer Science & Engineering, Washington University in St. Louis
2024	AAMAS Student Scholarship AAMAS Conference Committee — conference attendance and Doctoral Consortium
2014-2016	MHRD Scholarship Ministry of Human Resource Development, Government of India — for M.Tech. study

TECHNICAL SKILLS

SOFTWARE	Python, C/C++ , MATLAB; PyTorch, JAX, NumPy/SciPy, Hugging Face.
INFRASTRUCTURE	Distributed multi-GPU training, SLURM HPC clusters.

REFERENCES

Ph.D. advisor	Prof. Yevgeniy Vorobeychik Computer Science & Engineering, Washington University in St. Louis
Former advisor	Prof. Vineeth N. Balasubramanian Computer Science & Engineering, Indian Institute of Technology Hyderabad